WeiAn (Vivian) Huang
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EDUCATION

Master of Science in Business Analytics, *Questrom School of Business*, *Boston University* **Sept. 2024 - Jan. 2026**
Relative course: Machine Learning and Data Infrastructure in Health Care *Boston, MA*

Master of Health Administration, *National Yang Ming Chiao Tung University* **Sept. 2022 - Jun. 2024**
Relative course: Advanced Biostatistical Modeling and Data Analysis, Secondary Data Management and Analysis *Taipei, Taiwan*

Bachelor of Business Administration- Health Care Administration, *Taipei Medical University* **Sept. 2018 - Jun. 2022**
Relative course: Economics, Accounting, Management, Health Care Marketing *Taipei, Taiwan*

RELEVANT WORK EXPERIENCE

ProcDNA **Jul. 2025 – Aug. 2025**
Analytics Intern *Boston, MA*

- Supported oncology launch by performing data validation (QC) on provider datasets and aligning KPI definitions for client deliverables.
- Contributed to HIV predictive modeling by supporting feature engineering and developing HCP targeting frameworks.
- Delivered client-facing insights and organized a cross-office knowledge-sharing session (ProcTalk).

National Yang Ming Chiao Tung University
Research Analyst Assistant, Health Service Research Laboratory of Professor Nicole Huang **Sept. 2022 – Jul. 2024**

- Conducted large-scale healthcare claims-based analysis using national administrative datasets to evaluate program performance, utilization trends, and quality drivers.
- Built automated SAS/ SQL pipelines supporting cohort construction, longitudinal analysis across datasets covering ~23 million individuals.
- Applied statistical modeling (GEE), data validation, and quality checks to generate actionable insights; translated complex findings into clear summaries and presented results at EuHEA 2024.

Data Analyst, Medical Administrative Office **Jul. 2023 – Aug. 2023**

- Analyzed operational workflow data and identified process bottlenecks impacting capacity utilization; proposed data-driven process improvements improving workflow efficiency.
- Gained hands-on experience in hospital management, focusing on medical coding and quality control initiatives, while working across departments to improve workflow efficiency.
- Used Python (pandas) to analyze ED length of stay and discharge processes, applying field research, service blueprinting, and 5W1H to identify bottlenecks and optimize bed turnover for reducing ED crowding.

PROJECT EXPERIENCE

Forecasting ICU Demand and Readmission Risk Using MIMIC-IV **Sept. 2025 – Dec. 2025**
BA878: Machine Learning and Data Infrastructure in Health Care, Questrom School of Business, Boston University

- Built a decision-support analytics pipeline using MIMIC-IV datasets to predict 12-hour ICU admissions and 72-hour readmissions with clinical risk features.
- Analyzed large-scale clinical datasets using Python to identify drivers of ICU utilization, length of stay, and readmission risk, supporting operational planning.
- Designed a risk-based forecasting framework to support short-term ICU demand planning, proactive bed management, and discharge decision support.

GBH Investigations: Medical Debt Analytics Project **Sept. 2025 – Dec. 2025**
DS701: Tools for Data Science, Faculty of Computing & Data Sciences, Boston University — Collaboration with BU Spark!

- Cleaned and standardized 40k+ MassCourtsPlus civil cases using Python, correcting inconsistent case numbers, court names, and filing dates through custom parsing and fuzzy-matching techniques.
- Built a fuzzy keyword-mapping pipeline to classify cases into medical-related categories, improving classification accuracy.
- Analyzed plaintiff types, regional filing patterns, and medical-debt trends to support GBH investigative reporting.

Analyzing Yelp Review Trends for Strategic Restaurant Insights **Oct. 2024 – Dec. 2024**
BA775: Business Analytics Toolbox, Questrom School of Business, Boston University

- Analyzed 7M Yelp reviews using SQL and Tableau to explore drivers of customer satisfaction.
- Found accessibility linked to ~30% higher review volume and +0.5 rating increase, while elite users drove 10× more reviews and 35× higher engagement.
- Proposed strategies to boost engagement by 15–20%, including improving accessibility, and targeting elite reviewers.

Humana-Mays Healthcare Analytics Case Competition (Top 50) **Sept. 2024 – Oct. 2024**
Questrom School of Business, Boston University

- Analyzed data using Python from 1.5 million Medicare Advantage members, integrating demographics, claims, pharmacy, and behavioral data to predict disengagement from preventive care.
- Achieved 75.4% AUC using LightGBM with Bayesian and Randomized hyperparameter tuning; identified cost trends and digital engagement as top drivers of non-engagement.
- Designed an AI-supported care model featuring virtual consultations, case manager check-ins, and early financial interventions, projected to reduce care gaps by up to 25% for high-risk LPPO members.

SKILLS

Technical Skills: SQL, Python, SAS, Tableau, Power BI, BigQuery, Microsoft Office (Excel, Word, PowerPoint)